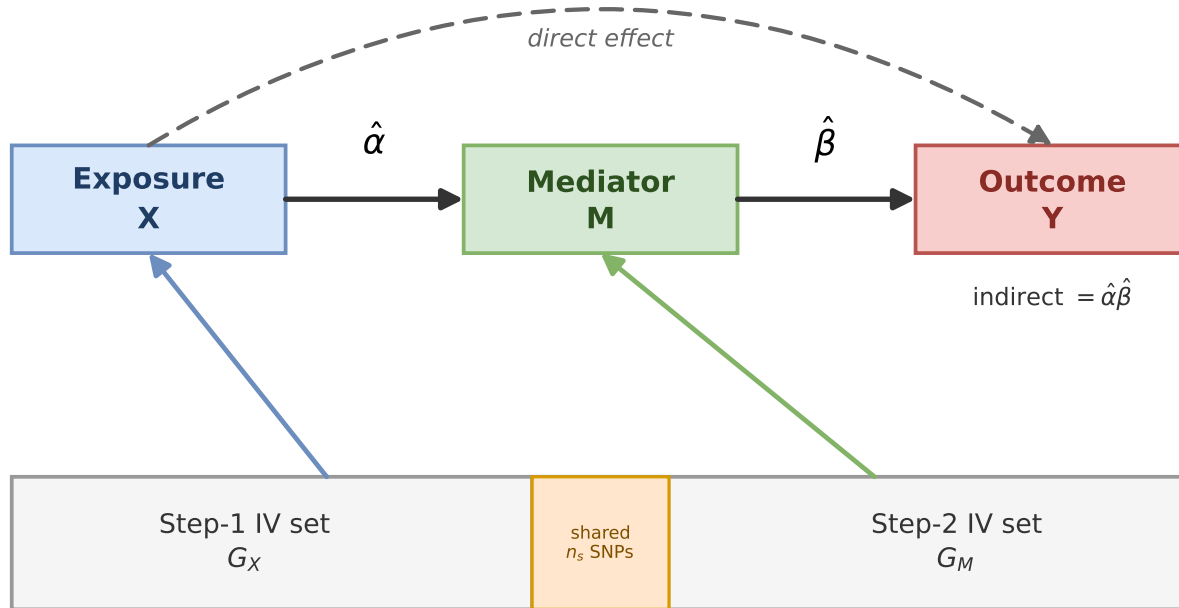


Figure 1. Conceptual diagram of two-step (product-method) MR mediation with overlapping instruments



$$\text{Cov}(\hat{\alpha}, \hat{\beta}) \neq 0 \text{ whenever } n_s > 0 \text{ or } \rho_{MY} \neq 0$$